

# The Central Electricity Regulatory Commission (Terms And Conditions For Tariff Determination From Renewable Energy Sources) Regulations, 2009

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Union of India - Act

The Central Electricity Regulatory Commission (Terms And Conditions For Tariff Determination From Renewable Energy Sources) Regulations, 2009

UNION OF INDIA

India

The Central Electricity Regulatory Commission (Terms And Conditions For Tariff Determination From Renewable Energy Sources) Regulations, 2009

Rule

THE-CENTRAL-ELECTRICITY-REGULATORY-COMMISSION-TERMS-AND-CONDITIONS-FOR-TARIFF-DETERMINATION-FROM-RENEWABLE-ENERGY-SOURCES-REGULATIONS-2009 of 2009

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The Central Electricity Regulatory Commission (Terms And Conditions For Tariff Determination From Renewable Energy Sources) Regulations, 2009

Published vide Notification No. L-7/186)201)/2009-CERC, dated 16.9.2009, published vide Notification Gazette of India,

Extraordinary, Part 3, Section 4, dated 17.9.2009.

9.

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In exercise of powers conferred under section 61 read with section 178(2)(s) of the Electricity Act, 2003 (36 of 2003

), and all other powers enabling it in this behalf, and after previous publication, the Central Electricity Regulatory Commission hereby makes the following regulations, namely:-

1.

Short title and commencement

-(1) These regulations may be called the Central Electricity Regulatory Commission (Terms and Conditions for Tariff determination from Renewable Energy Sources) Regulations, 2009.

(2)

These regulations shall come into force from the date of their publication in the Official Gazette.

2.

Definitions and Interpretation

-(1) In these regulations, unless the context otherwise requires,-

(a)

"Act" means the Electricity Act, 2003 (

36 of 2003

);

(b)

"Auxiliary energy consumption" or "AUX" in relation to a period in case of a generating station means the quantum of energy consumed by auxiliary equipment of the generating station, and transformer losses within the generating station, expressed as a percentage of the sum of gross energy generated at the generator terminals of all the units of the generating station;

(c)

"Biomass" means wastes produced during agricultural and forestry operations (for example straws and stalks) or produced as a by-product of processing operations of agricultural produce (e.g., husks, shells, deoiled cakes, etc.); wood produced in dedicated energy plantations or recovered from wild bushes/weeds; and the wood waste produced in some industrial operations;

(d)

"Capital Cost" means the capital cost as defined in regulations 12, 24, 28, 34, 47, 57 and 61;

(e)

"Commission" means the Central Electricity Regulatory Commission referred to in sub-section (1) of section 76 of the Act;

(f)

"Conduct of Business Regulations" means the Central Electricity Regulatory Commission (Conduct of Business) Regulations, 1999 as amended from time to time;

(g)

"Control Period or Review Period" means the period during which the norms for determination of tariff specified in these regulations shall remain valid;

(h)

"Gross calorific value" or "GCV" in relation to a fuel used in generating station means the heat produced in kCal by complete combustion of one kilogram of solid fuel or one litre of liquid fuel or one standard cubic meter of gaseous fuel, as the case may be;

(i)

"Gross station heat rate" or "GHR" means the heat energy input in kCal required to generate one kWh of electrical energy at generator terminals of a thermal generating station;

(j)

"Hybrid Solar Thermal Power Plant" means the solar thermal power plant that uses other forms of energy input sources alongwith solar thermal energy for electricity generation, and wherein not less than 75% of electricity is generated from

solar energy component;

(k)

"Installed capacity" or "IC" means the summation of the name plate capacities of all the units of the generating station or the capacity of the generating station (reckoned at the generator terminals), approved by the Commission from time to time;

(l)

"Inter-connection Point" shall mean interface point of renewable energy generating facility with the transmission system or distribution system, as the case may be:

(a)

in relation to wind energy projects and Solar Photovoltaic Projects, inter-connection point shall be line isolater on outgoing feeder on HV side of the pooling sub-station;

(b)

in relation to small hydro power, biomass power and non fossil fuel based cogeneration power projects and Solar Thermal Power Projects the, inter-connection point shall be line isolator on outgoing feeder on HV side of generator transformer;

(m)

"MNRE" means the Ministry of New and Renewable Energy of the Government of India;

(n)

"Non-firm power" means the power generated from renewable sources, the hourly variation of which is dependent upon nature's phenomenon like sun, cloud, wind, etc., that cannot be accurately predicted;

(o)

"Non-fossil fuel based co-generation" means the process in which more than one form of energy (such as steam and electricity) are produced in a sequential manner by use of biomass provided the project may qualify to be a co-generation project if it fulfils the eligibility criteria as specified in clause (4) of regulation 4;

(p)

"Operation and maintenance expenses" or "O&M expenses" means the expenditure incurred on operation and maintenance of the project, or part thereof, and includes the expenditure on manpower, repairs, spares, consumables, insurance and overheads;

(q)

"Project" means a generating station or the evacuation system upto inter-connection point, as the case may be, and in case of a small hydro generating station includes all components of generating facility such as dam, intake water conductor system, power generating station and generating units of the scheme, as apportioned to power generation;

(r)

"Renewable Energy" means the grid quality electricity generated from renewable energy sources;

(s)

"Renewable Energy Power Plants" means the power plants other than the conventional power plants generating grid quality electricity from renewable energy sources;

(t)

"Renewable Energy Sources" means renewable sources such as small hydro, wind, solar including its integration with combined cycle, biomass, bio fuel cogeneration, urban or municipal waste and other such sources as approved by the MNRE;

(u)

"Small Hydro" means Hydro Power projects with a station capacity up to and including 25 MW;

(v)

"Solar PV power" means the Solar Photo Voltaic power project that uses sunlight for direct conversion into electricity through Photo Voltaic technology;

(w)

"Solar Thermal power" means the Solar Thermal power project that uses sunlight for direct conversion into electricity through Concentrated Solar Power technology based on either line focus or point focus principle;

(x)

"Tariff Period" means the period for which tariff is to be determined by the Commission on the basis of norms specified

under these regulations;

(y)

"Useful Life" in relation to a unit of a generating station including evacuation system shall mean the following duration from the date of commercial operation (COD) of such generation facility, namely:-

(a)

Wind energy power project - 25 years

(b)

Biomass power project, non-fossil fuel cogeneration - 20 years

(c)

Small Hydro Plant - 35 years

(d)

Solar PV/Solar thermal power plants - 25 years;

(z)

"Year" means a financial year.

(2)

Save as aforesaid and unless repugnant to the context or if the subject-matter otherwise requires, words and expressions used in these regulations and not defined, but defined in the Act, or the Indian Electricity Grid Code or the Central Electricity Regulatory Commission (Terms and Conditions of Tariff) Regulations, 2009 shall have the meanings assigned to them respectively in the Act or the Indian Electricity Grid Code or the Central Electricity Regulatory Commission (Terms and Conditions of Tariff) Regulations, 2009.

3.

Scope and extent of application

.-These regulations shall apply in all cases where tariff, for a generating station or a unit thereof based on renewable sources of energy, is to be determined by the Commission under section 62 read with section 79 of the Act:

Provided that in cases of wind, small hydro projects, biomass power, non-fossil fuel based cogeneration projects, solar PV and Solar Thermal power projects, these regulations shall apply subject to the fulfilment of eligibility criteria specified in regulation 4 of these regulations.

4.

Eligibility Criteria

.-

(1)

Wind power project.-Located at the wind sites having minimum annual mean Wind Power Density (WPD) of 200 Watt/m<sup>2</sup> measured at hub height of 50 meters and using new wind turbine generators.

(2)

Small hydro project.-Located at the sites approved by State Nodal Agency/State Government using new plant and machinery, and installed power plant capacity to be lower than or equal to 25 MW at single location.

(3)

Biomass power project.-Biomass power projects using new plant and machinery based on Rankine Cycle technology and using biomass fuel sources, provided use of fossil fuel is restricted only to 15% of total fuel consumption on annual basis.

(4)

Non-fossil fuel based co-generation project.-The project shall qualify to be termed as a non-fossil fuel based co-generation project, if it is using new plant and machinery and is in accordance with the definition and also meets the qualifying requirement outlined below:

Topping cycle mode of co-generation.-Any facility that uses non-fossil fuel input for the power generation and also utilizes the thermal energy generated for useful heat applications in other industrial activities simultaneously:

Provided that for the co-generation facility to qualify under topping cycle mode, the sum of useful power output and one half the useful thermal output be greater than 45% of the facility's energy consumption, during season."

Explanation.-For the purposes of this clause,-

(a)

"Useful power output" is the gross electrical output from the generator. There will be an auxiliary consumption in the

cogeneration plant itself (e.g., the boiler feed pump and the FD/ID fans). In order to compute the net power output it would be necessary to subtract the auxiliary consumption from the gross output. For simplicity of calculation, the useful power output is defined as the gross electricity (kWh) output from the generator.

(b)

"Useful Thermal Output" is the useful heat (steam) that is provided to the process by the cogeneration facility.

(c)

"Energy Consumption" of the facility is the useful energy input that is supplied by the fuel (normally bagasse or other such biomass fuel).

(5)

Solar PV and Solar Thermal Power Projects.-Based on Technologies approved by MNRE.

## Chapter 1

### General Principles

5.

#### Control Period or Review Period

.-The Control Period or Review Period under these regulations shall be of three years, of which the first year shall be the period from the date of notification of these regulations to 31-3-2010:

Provided that the benchmark capital cost for Solar PV and Solar thermal projects may be reviewed annually by the Commission:

Provided further that the tariff determined as per these regulations for the RE projects commissioned during the Control Period, shall continue to be applicable for the entire duration of the Tariff Period as specified in regulation 6 below:

Provided also that the revision in regulations for next Control Period shall be undertaken at least six months prior to the end of the first Control Period and in case regulations for the next Control Period are not notified until commencement of next Control Period, the tariff norms as per these regulations shall continue to remain applicable until notification of the revised regulations subject to adjustments as per revised regulations.

6.

#### Tariff Period

.- (1) The Tariff Period for Renewable Energy power projects except in case of Small hydro projects below 5 MW, Solar PV, and Solar thermal power projects shall be thirteen (13) years.

(2)

In case of small hydro projects below 5 MW, the Tariff Period shall be thirty five (35) years.

(3)

In case of Solar PV and Solar thermal power projects the Tariff Period shall be twenty five years (25) years.

(4)

Tariff Period under these regulations shall be considered from the date of commercial operation of the renewable energy generating stations.

(5)

Tariff determined as per these regulations shall be applicable for Renewable Energy power projects, only for the duration of the Tariff Period as stipulated under regulation 6(1), (2) and (3).

7.

#### Project specific tariff

.- (1) Project specific tariff, on case to case basis, shall be determined by the Commission for the following types of projects:

(a)

Municipal Solid Waste Projects;

(b)

any other new renewable energy technologies approved by MNRE;

(c)

the renewable energy projects which have been commissioned before the notification of these regulations but for which no power purchase agreement has been signed until the date of notification of these regulations;

(d)

Solar PV and Solar Thermal Power projects, if a project developer opts for project specific tariff:

Provided that the Commission while determining the project specific tariff for Solar PV and Solar Thermal shall be guided by the provisions of Chapters 8 of these regulations;

(e)

Hybrid Solar Thermal Power plants;

(f)

Biomass project other than that based on Rankine Cycle technology application with water cooled condenser.

(2)

Determination of project specific tariff for generation of electricity from such renewable energy sources shall be in accordance with such terms and conditions as stipulated under relevant orders of the Commission:

Provided that the financial norms as specified under Chapter 2 of these regulations, except for capital cost, shall be ceiling norms while determining the project specific tariff.

8.

Petition and proceedings for determination of tariff

-(1) The Commission shall determine the generic tariff on the basis of suo motu petition at least six months in advance at the beginning of each year of the Control Period for renewable energy technologies for which norms have been specified under the regulations.

(2)

A petition for determination of project specific tariff shall be accompanied by such fee as may be determined by regulations and shall be accompanied by-

(a)

information in forms 1.1, 1.2, 2.1 and 2.2 as the case may be, and as appended in these regulations;

(b)

detailed project report outlining technical and operational details, site specific aspects, premise for capital cost and financing plan, etc.;

(c)

a statement of all applicable terms and conditions and expected expenditure for the period for which tariff is to be determined;

(d)

a statement containing full details of calculation of any subsidy and incentive received, due or assumed to be due from the Central Government and/or State Government. This statement shall also include the proposed tariff calculated without consideration of the subsidy and incentive;

(e)

any other information that the Commission requires the petitioner to submit.

(3)

The proceedings for determination of tariff shall be in accordance with the Conduct of Business Regulations.

9. Tariff Structure

-(1) The tariff for renewable energy technologies shall be single part tariff consisting of the following fixed cost components:

(a)

Return on equity;

(b)

Interest on loan capital;

(c)

Depreciation;

(d)

Interest on working capital;

(e)

Operation and maintenance expenses:

Provided that for renewable energy technologies having fuel cost component, like biomass power projects and non-fossil fuel based cogeneration, single part tariff with two components, fixed cost component and fuel cost component, shall be determined.

10.

#### Tariff Design

-(1) The generic tariff shall be determined on levelled basis for the Tariff Period:

Provided that for renewable energy technologies having single part tariff with two components, tariff shall be determined on levelled basis considering the year of commissioning of the project for fixed cost component while the fuel cost component shall be specified on year of operation basis.

(2)

For the purpose of levelled tariff computation, the discount factor equivalent to weighted average cost of capital shall be considered.

(3)

Levelled tariff shall be carried out for the "useful life" of the Renewable Energy project while Tariff shall be specified for the period equivalent to "Tariff Period".

11.

#### Despatch principles for electricity generated from Renewable Energy Sources

-(1) All renewable energy power plants except for biomass power plants with installed capacity of 10 MW and above, and non-fossil fuel based cogeneration plants shall be treated as "MUST RUN" power plants and shall not be subjected to "merit order despatch" principles.

(2)

The biomass power generating station with an installed capacity of 10 MW and above and non-fossil fuel based co-generation projects shall be subjected to scheduling and despatch code as specified under Indian Electricity Grid Code (IEGC) and Central Electricity Regulatory Commission (Unscheduled Interchange and Related Matters) Regulations, 2009 including amendments thereto.

#### Chapter 2

#### Financial Principles

12.

#### Capital Cost

-.The norms for the Capital Cost as specified in the subsequent technology specific chapters shall be inclusive of all capital work including plant and machinery, civil work, erection and commissioning, financing and interest during construction, and evacuation infrastructure up to inter-connection point:

Provided that for project specific tariff determination, the generating company shall submit the break-up of capital cost items along with its petition in the manner specified under regulation 8.

13.

#### Debt Equity Ratio

-(1) For generic tariff to be determined based on suo motu petition, the debt equity ratio shall be 70 : 30.

(2)

For Project specific tariff, the following provisions shall apply:-

If the equity actually deployed is more than 30% of the capital cost, equity in excess of 30% shall be treated as normative loan:

Provided that where equity actually deployed is less than 30% of the capital cost, the actual equity shall be considered for determination of tariff:

Provided further that the equity invested in foreign currency shall be designated in Indian rupees on the date of each investment.

14.

#### Loan and Finance Charges

-(1) Loan tenure.-For the purpose of determination of tariff, loan tenure of 10 years shall be considered.

(2)

Interest rate.-(a) The loans arrived at in the manner indicated above shall be considered as gross normative loan for calculation for interest on loan. The normative loan outstanding as on April 1st of every year shall be worked out by deducting the cumulative repayment up to March 31st of previous year from the gross normative loan.

(b)

For the purpose of computation of tariff, the normative interest rate shall be considered as average long term prime

lending rate (LTPLR) of State Bank of India (SBI) prevalent during the previous year plus 150 basis points.

(c)

Notwithstanding any moratorium period availed by the generating company, the repayment of loan shall be considered from the first year of commercial operation of the project and shall be equal to the annual depreciation allowed.

15.

Depreciation

-(1) The value base for the purpose of depreciation shall be the Capital Cost of the asset admitted by the Commission. The Salvage value of the asset shall be considered as 10% and depreciation shall be allowed up to maximum of 90% of the Capital Cost of the asset.

(2)

Depreciation per annum shall be based on "Differential Depreciation Approach" over loan tenure and period beyond loan tenure over useful life computed on "Straight Line Method". The depreciation rate for the first 10 years of the Tariff Period shall be 7% per annum and the remaining depreciation shall be spread over the remaining useful life of the project from 11th year onwards.

(3)

Depreciation shall be chargeable from the first year of commercial operation:

Provided that in case of commercial operation of the asset for part of the year, depreciation shall be charged on pro rata basis.

16.

Return on Equity

-(1) The value base for the equity shall be 30% of the capital cost or actual equity (in case of project specific tariff determination) as determined under regulation 13.

(2)

The normative return on equity shall be:-

(a)

Pre-tax 19% per annum for the first 10 years.

(b)

Pre-tax 24% per annum 11th years onwards.

17.

Interest on Working Capital

-(1) The working capital requirement in respect of wind energy projects, small hydro power, solar PV and Solar thermal power projects shall be computed in accordance with the following:-

Wind Energy/Small Hydro Power/Solar PV/Solar thermal

(a)

Operation & Maintenance expenses for one month;

(b)

Receivables equivalent to 2 (Two) months of energy charges for sale of electricity calculated on the normative CUF;

(c)

Maintenance spare @ 15% of operation and maintenance expenses

(2)

The Working Capital requirement in respect of biomass power projects and non-fossil fuel based co-generation projects shall be computed in accordance with the following clause:

Biomass Power and Non-fossil fuel Co-generation

(a)

Fuel costs for four months equivalent to normative PLF;

(b)

Operation and Maintenance expense for one month;

(c)

Receivables equivalent to 2 (Two) months of fixed and variable charges for sale of electricity calculated on the target PLF;

(d)



Maintenance spare @ 15% of operation and maintenance expenses

(3)

Interest on Working Capital shall be at interest rate equivalent to average State Bank of India short term PLR during the previous year plus 100 basis points.

18.

Operation and Maintenance Expenses

-(1) "Operation and Maintenance or O&M expenses" shall comprise repair and maintenance (R&M), establishment including employee expenses, and administrative and general expenses.

(2)

Operation and maintenance expenses shall be determined for the Tariff Period based on normative O&M expenses specified by the Commission subsequently in these regulations for the first Year of Control Period.

(3)

Normative O&M expenses allowed during first year of the Control Period (i.e., FY 2009-10) under these regulations shall be escalated at the rate of 5.72% per annum over the Tariff Period.

19.

Rebate

-(1) For payment of bills of the generating company through letter of credit, a rebate of 2% shall be allowed.

(2)

Where payments are made other than through letter of credit within a period of one month of presentation of bills by the generating company, a rebate of 1% shall be allowed.

20.

Late payment surcharge

.-In case the payment of any bill for charges payable under these regulations is delayed beyond a period of 60 days from the date of billing, a late payment surcharge at the rate of 1.25% per month shall be levied by the generating company.

21.

Sharing of CDM Benefits

.-The proceeds of carbon credit from approved CDM project shall be shared between generating company and concerned beneficiaries in the following manner, namely:-

(a)

100% of the gross proceeds on account of CDM benefit to be retained by the project developer in the first year after the date of commercial operation of the generating station;

(b)

In the second year, the share of the beneficiaries shall be 10% which shall be progressively increased by 10% every year till it reaches 50%, where after the proceeds shall be shared in equal proportion, by the generating company and the beneficiaries.

22.

Subsidy or incentive by the Central/State Government

.-The Commission shall take into consideration any incentive or subsidy offered by the Central or State Government, including accelerated depreciation benefit if availed by the generating company, for the renewable energy power plants while determining the tariff under these regulations:

Provided that the following principles shall be considered for ascertaining income tax benefit on account of accelerated depreciation, if availed, for the purpose of tariff determination:-

(i)

Assessment of benefit shall be based on normative capital cost, accelerated depreciation rate as per relevant provisions under Income-Tax Act and corporate income-tax rate.

(ii)

Capitalisation of RE projects during second half of the fiscal year.

Per unit benefit shall be derived on levellised basis at discount factor equivalent to weighted average cost of capital.

23.

Taxes and Duties

.-Tariff determined under these regulations shall be exclusive of taxes and duties as may be levied by the appropriate Government:

Provided that the taxes and duties levied by the appropriate Government shall be allowed as pass through on actual incurred basis.

### Chapter 3

#### Technology Specific Parameters For Wind Energy

24.

##### Capital Cost

.- (1) The capital cost for wind energy project shall include Wind turbine generator including its auxiliaries, land cost, site development charges and other civil works, transportation charges, evacuation cost up to inter-connection point, financing charges and IDC.

(2)

The capital cost for wind energy projects shall be Rs. 515 Lakhs/MW (FY 2009-10 during first year of Control Period) and shall be linked to indexation formula as outlined under regulation 25.

25.

##### Capital Cost Indexation Mechanism

.-The following indexation mechanism shall be applicable in case of wind energy projects for adjustments in capital cost over the Control Period with the changes in Wholesale Price Index for Steel and Electrical Machinery:

$$CC(n) = P\&M(n) * (1+F1+F2+F3)$$

$$P\&M(n) = P\&M(0) * (1+d(n))$$

$$d(n) = [a * \{(SI(n-1)/SI(0)-1) + \{(EI(n-1)/EI(0))-1\}]/(a+b)$$

Where,-

CC(n) = Capital Cost for nth year

P&M(n) = Plant and Machinery Cost for nth year

P&M(0) = Plant and Machinery Cost for the base year

Note.- P&M(0) is to computed by

dividing the base capital cost (for the first year of the Control Period) by (1+F1+F2+F3), i.e., Rs. 515 Lakh per MW/1.25 = Rs. 412 Lakh per MW.

d(n) = Capital Cost escalation factor for year

(n)

of Control period.

SI(n-1) = Average WPI Steel Index prevalent for calendar year (n-1) of the Control Period.

SI(0) - Average WPI Steel Index prevalent for calendar year (0) at the beginning of the Control Period, i.e., January 2008.

to December 2008.

EI(n-1) = Average WPI Electrical Machinery Index prevalent for calendar year (n-1) of the Control Period

EI(0) = Average WPI Electrical and Machinery Index prevalent for calendar year (0) at the beginning of the Control Period, i.e., January 2008 to December 2008

a = Constant to be determined by Commission from time to time,

(In default it is 0.6), for weight age to Steel Index

b = Constant to be determined by Commission from time to time,

(In default if its 0.4), for weight age to Electrical Machinery Index

F1 = Factor for Land and Civil Work (0.08)

F2 = Factor for Erection and Commissioning (0.07)

F3 = Factor for IDC and Financing Cost (0.10)

26.

Capacity Utilisation Factor

-(1) CUF norms for this Control Period shall be as follows:-

Annual Mean Wind Power Density (W /m<sup>2</sup>)

CUF

200-250

20%

250-300

23%

300-400

27%

> 400

30%

(2)

The annual mean wind power density specified in sub-regulation (1) above shall be measured at 50 meter hub-height.

(3)

For the purpose of classification of wind energy project into particular wind zone class, the State-wise wind power density map prepared by Centre for Wind Energy Technology (C-WET) and enclosed as Schedule to these regulations, shall be considered:

Provided that the Commission may by notification in Official Gazette, amend the schedule from time to time, based on the input provided by C-WET/MNRE.

27.

Operation and Maintenance Expenses

-(1) Normative O&M expenses for the first year of the Control Period (i.e., FY 2009-10) shall be Rs. 6.50 Lakh per MW.

(2)

Normative O&M expenses allowed under these regulations shall be escalated at the rate of 5.72% per annum over the Tariff Period to compute the levelled tariff.

Chapter 4

Technology Specific Parameters For Small Hydro Project

28.

Capital Cost

-(1) The normative capital cost for small hydro projects during first year of Control Period (FY 2009-10) shall be as follows:-

Region

Project Size

Capital Cost

(Rs Lakh/MW)

Himanchal Pradesh, Uttarkhand and North Eastern States

Below 5 MW

5 MW to 25 MW

700

630

Other States

Below 5 MW

5 MW to 25 MW

550

500

(2)

The capital cost for subsequent years shall be determined on the basis of indexation formula as outlined under regulation 29"> (2) The capital cost for subsequent years shall be determined on the basis of indexation formula as

outlined under regulation 29.

29.

Capital Cost Indexation Mechanism

.-The following indexation mechanism shall be applicable in case of small hydro power projects for adjustments in capital cost over the Control Period with the changes in Wholesale Price Index for Steel and Electrical Machinery:

$$CC(n) = P\&M(n) * (1+F1+F2+F3)$$

$$P\&M(n) = P\&M(0) * (1+d(n))$$

$$d(n) = [a*\{(SI(n-1)/SI(0)-1)\}+\{(EI(n-1)/EI(0))-1\}]/(a+b)$$

Where,-

CC(n) = Capital Cost for nth year

P&M(n) = Plant and Machinery Cost for nth year

P&M(0) = Plant and Machinery Cost for the base year

Note.- P&M(0) is to computed by dividing the base capital cost (for the first year of the Control Period) by (1+F1+F2+F3), i.e.

Small Hydro Project

Base Capital

Cost (Rs. Lakh/MW)

Factor

(1+F1+F2+F3)

P&M(0)`

(Rs. Lakh/MW)

SHP(< 5 MW) (HP, Uttarakhand, NE States)

700

1.40

500

SHP (5-25 MW) (HP, Uttarakhand, NE States)

630

1.40

450

SHP (<5 MW) (other States)

550

1.40

393

SHP (5-25 MW) (other States)

500

1.40

357

D(n) = Capital Cost escalation factor for year (n) of Control period.

SI(n-1) = Average WPI Steel Index prevalent for calendar year (n-1) of the Control Period.

SI(0) - Average WPI Steel Index prevalent for calendar year (0) at the beginning of the Control Period, i.e., January 2008

EI(n-1) = Average WPI Electrical Machinery Index prevalent for calendar year (n-1) of the Control Period

EI(0) = Average WPI Electrical and Machinery Index prevalent for calendar year (0) at the beginning of the Control Period, i.e., January 2008 to December 2008

a = Constant to be determined by Commission from time to

- time,  
(In default it is 0.6), for weightage to Steel Index  
b = Constant to be determined by Commission from time to time,  
(In default if its 0.4), for weightage to Electrical Machinery Index  
F1 = Factor for Land and Civil Work (0.16)  
F2 = Factor for Erection and Commissioning (0.10)  
F3 = Factor for IDC and Financing Cost (0.14)
30.  
Capacity Utilisation Factor  
.-Capacity Utilisation factor for the small hydro projects located in Himanchal Pradesh, Uttarakhand and North Eastern States shall be 45% and for other States, CUF shall be 30%.  
Explanation.-For the purpose of this regulation normative CUF is net of free power to the home state if any, and any quantum of free power if committed by the developer over and above the normative CUF shall not be factored into the tariff.
31.  
Auxiliary Consumption  
.-Normative Auxiliary Consumption for the small hydro projects shall be 1.0%.
32.  
Operation and Maintenance Expenses  
.- (1) Normative O&M expenses for the first year of the Control Period (i.e. FY 2009-10) shall be as follows:
- | Region                                                  | Project Size  | O&M Expense<br>(Rs Lakh/MW) |
|---------------------------------------------------------|---------------|-----------------------------|
| Himanchal Pradesh, Uttarakhand and North Eastern States | Below 5 MW    | 21                          |
|                                                         | 5 MW to 25 MW | 15                          |
| Other States                                            | Below 5 MW    | 17                          |
|                                                         | 5 MW to 25 MW | 12                          |
- (2)  
Normative O&M expenses allowed under these regulations shall be escalated at the rate of 5.72% per annum for the Tariff Period for the purpose of determination of levellised tariff.
- Chapter 5  
Technology Specific Parameters For Biomass Based Power Projects
33.  
Technology Aspect  
.-The norms for tariff determination specified hereunder are for biomass power projects based on Rankine cycle technology application using water cooled condenser.
34.  
Capital Cost  
.-The normative capital cost for the biomass power projects shall be Rs. 450 Lakh/MW (FY 2009-10 during first year of Control Period) and shall be linked to indexation formula as outlined under regulation 35.
35.  
Capital Cost Indexation Mechanism  
.-The following indexation mechanism shall be applicable in case of biomass power projects for adjustment in capital

cost over the Control Period with the changes in Wholesale Price Index for Steel and Electrical Machinery:

$$CC(n) = P\&M(n) * (1+F1+F2+F3$$

$$P\&M(n) = P\&M(0) * (1+d(n)$$

$$d(n) = [a*\{(SI(n-1)/SI(0))-1\}+b*\{(EI(n-1)/EI(0))-1\}]/(a+b)$$

Where,-

CC(n) = Capital Cost for nth year

P&M(n) = Plant and Machinery Cost for nth year

P&M(0) = Plant and Machinery Cost for the base year

Note. - P&M(0) is to be computed by

dividing the base capital cost (for the first year of the Control Period) by

(1+F1+F2+F3), i.e., Rs. 450 Lakh per MW/1.33 =

Rs. 338 Lakh per MW.

d(n) = Capital Cost escalation factor for year

(n)

of Control Period.

SI(n-1) = Average WPI Steel Index prevalent for calendar

year (n-1) of the Control Period.

SI(0) = Average WPI Steel Index prevalent for

calendar year (0) at the beginning of the Control Period, i.e., January 2008 to December 2008

EI(n-1) = Average WPI Electrical Machinery Index

prevalent for calendar year (n-1) of the Control Period.

EI(0) = Average WPI Electrical and Machinery Index prevalent for calendar year (0) at the beginning of the Control Period, i.e., January 2008 to December 2008

a = Constant to be determined by Commission from time to time,

b = Constant to be determined by Commission from time to time,

(In default it is 0.3), for weightages to Electrical Machinery Index

F1 = Factor for Land and Civil Works (0.10)

F2 = Factor for Erection and Commissioning (0.09)

F3 = Factor for IDC and Financing Cost (0.14)

36.

Plant Load Factor

-(1) Threshold Plant Load Factor for determining fixed charge component of Tariff shall be:

1.

During stabilization:

60%

2.

During the remaining period of the first year (after stabilization):

70%

3.

From 2nd year onwards:

80%

37.

Auxiliary Consumption

.-The auxiliary power consumption factor shall be 10% for the determination of tariff.

38.

Station Heat Rate

.-The Station Heat Rate for biomass power projects shall be 3800 kCal/kWh.

39.

Operation and Maintenance Expenses

-(1) Normative O&M expenses for the first year of the Control Period (i.e., FY 2009-10) shall be Rs. 20.25 Lakh per MW.

(2)

Normative O&M expenses allowed at the commencement of the Control Period (i.e., FY 2009-10) under these regulations shall be escalated at the rate of 5.72% per annum.

40.

Fuel Mix

-(1) The biomass power plant shall be designed in such a way that it uses different types of non-fossil fuels available within the vicinity of biomass power project such as crop residues, agro-industrial residues, forest residues, etc., and other biomass fuels as may be approved by MNRE.

(2)

The Biomass Power Generating Companies shall ensure fuel management plan to ensure adequate availability of fuel to meet the respective project requirements.

41.

Use of Fossil Fuel

-.The use of fossil fuels shall be limited to the extent of 15% of total fuel consumption on annual basis.

42.

Monitoring Mechanism for the use of fossil fuel

-(1) The Project developer shall furnish a monthly fuel usage statement and monthly fuel procurement statement duly certified by Chartered Accountant to the beneficiary (with a copy to appropriate agency appointed by the Commission for the purpose of monitoring the fossil and non-fossil fuel consumption) for each month, along with the monthly energy bill. The statement shall cover details such as-

(a)

Quantity of fuel (in tonnes) for each fuel type (biomass fuels and fossil fuels) consumed and procured during the month for power generation purposes,

(b)

Cumulative quantity (in tonnes) of each fuel type consumed and procured till the end of that month during the year,

(c)

Actual (gross and net) energy generation (denominated in units) during the month,

(d)

Cumulative actual (gross and net) energy generation (denominated in units) until the end of that month during the year,

(e)

Opening fuel stock quantity (in tonnes),

(f)

Receipt of fuel quantity (in tonnes) at the power plant site, and

(g)

Closing fuel stock quantity (in tonnes) for each fuel type (biomass fuels and fossil fuels) available at the power plant site.

(2)

Non-compliance with the condition of fossil fuel usage by the project developer, during any financial year, shall result in withdrawal of applicability of tariff as per these regulations for such biomass based power project.

43.

Calorific Value

-.The Calorific Value of the biomass fuel used for the purpose of determination of tariff shall be as follows:

State

Calorific Value

(kCal/kg)

Andhra Pradesh

3275

Haryana

|                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3458                                                                                                                                                                                                                                                                                                                                                                               |
| Maharashtra                                                                                                                                                                                                                                                                                                                                                                        |
| 3611                                                                                                                                                                                                                                                                                                                                                                               |
| Madhya Pradesh                                                                                                                                                                                                                                                                                                                                                                     |
| 3612                                                                                                                                                                                                                                                                                                                                                                               |
| Punjab                                                                                                                                                                                                                                                                                                                                                                             |
| 3368                                                                                                                                                                                                                                                                                                                                                                               |
| Rajasthan                                                                                                                                                                                                                                                                                                                                                                          |
| 3689                                                                                                                                                                                                                                                                                                                                                                               |
| Tamil Nadu                                                                                                                                                                                                                                                                                                                                                                         |
| 3300                                                                                                                                                                                                                                                                                                                                                                               |
| Uttar Pradesh                                                                                                                                                                                                                                                                                                                                                                      |
| 3371                                                                                                                                                                                                                                                                                                                                                                               |
| Other States                                                                                                                                                                                                                                                                                                                                                                       |
| 3467                                                                                                                                                                                                                                                                                                                                                                               |
| 44.                                                                                                                                                                                                                                                                                                                                                                                |
| Fuel Cost                                                                                                                                                                                                                                                                                                                                                                          |
| .-Biomass fuel price during first year of the Control Period (i.e., FY 2009-10) shall be as presented in the table below and shall be linked to index formulae as specified under regulation 45. Alternatively, for each subsequent year of the Tariff Period, the normative escalation factor of 5% per annum shall be applicable at the option of the biomass project developer. |
| State                                                                                                                                                                                                                                                                                                                                                                              |
| Biomass Price                                                                                                                                                                                                                                                                                                                                                                      |
| (Rs /MT)                                                                                                                                                                                                                                                                                                                                                                           |
| Andhra Pradesh                                                                                                                                                                                                                                                                                                                                                                     |
| 1301                                                                                                                                                                                                                                                                                                                                                                               |
| Haryana                                                                                                                                                                                                                                                                                                                                                                            |
| 2168                                                                                                                                                                                                                                                                                                                                                                               |
| Maharashtra                                                                                                                                                                                                                                                                                                                                                                        |
| 1801                                                                                                                                                                                                                                                                                                                                                                               |
| Madhya Pradesh                                                                                                                                                                                                                                                                                                                                                                     |
| 1299                                                                                                                                                                                                                                                                                                                                                                               |
| Punjab                                                                                                                                                                                                                                                                                                                                                                             |
| 2092                                                                                                                                                                                                                                                                                                                                                                               |
| Rajasthan                                                                                                                                                                                                                                                                                                                                                                          |
| 1822                                                                                                                                                                                                                                                                                                                                                                               |
| Tamil Nadu                                                                                                                                                                                                                                                                                                                                                                         |
| 1823                                                                                                                                                                                                                                                                                                                                                                               |
| Uttar Pradesh                                                                                                                                                                                                                                                                                                                                                                      |
| 1518                                                                                                                                                                                                                                                                                                                                                                               |
| Other States                                                                                                                                                                                                                                                                                                                                                                       |
| 1797                                                                                                                                                                                                                                                                                                                                                                               |
| 45.                                                                                                                                                                                                                                                                                                                                                                                |
| Fuel Price Indexation Mechanism                                                                                                                                                                                                                                                                                                                                                    |
| .- (1) In case of biomass power projects, the following indexing mechanism for adjustment of fuel prices for each year of operation will be applicable for determination of applicable variable charge component of tariff, in case developer wishes to opt for indexing mechanism:                                                                                                |
| $P(n) = P(n-1) * \{a * (WPI(n)/WPI(n-1)) + b * (1 + IRC)(n-1) + c * (Pd(n)/Pd(n-1))\}$                                                                                                                                                                                                                                                                                             |
| Where,-                                                                                                                                                                                                                                                                                                                                                                            |
| P(n) = Price per ton of biomass for the nth year                                                                                                                                                                                                                                                                                                                                   |



to be considered for tariff determination

$P(n-1)$  = Price per ton of biomass for the (n-1)th

year to be considered for tariff determination.  $P_1$  shall be Biomass price for FY 2009-10 as specified under regulation 44.

$a$  = Factor representing fuel handling cost

$b$  = Factor representing fuel cost

$c$  = Factor representing transportation cost

$IRC(n-1)$  = Average Annual Inflation Rate for

indexed energy charge component in case of captive coal mine source (in %) to

be applicable for (n-1)th year, as may be specified by CERC for

"Payment purpose" as per competitive Bidding Guidelines

$P_{dn}$  = Weighted average price of HSD for nth year.

$P_{dn-1}$  = Weighted average price of HSD for (n-1)th year.

$WPI_n$  = Whole sale price index for the month of April of nth year

$WPI_{n-1}$  = Wholesale price index for month of April of (n-1)th year.

Where  $a$ ,  $b$  &  $c$  will be specified by the Commission from time to time. In default, these factors shall be 0.2, 0.6 & 0.2 respectively.

(2)

Variable Charge for the nth year shall be determined as under :

i.e.  $VC_n = VC_1 \times (P_n/P_1)$

or  $VC_n = VC_1 \times (1.05)^{(n-1)}$  (optional)

where,

$VC_1$  represents the Variable Charge based on Biomass Price  $P$  for FY 2009-10 as specified under regulation 44 and shall be

determined as under:

$VC_1 =$

Station Heat Rate

(SHR)

$\times$

1

$\times$

$P_1$

Gross Calorific Value (GCV)

(1-Aux Consum.

Factor)

1000

Chapter 6

Technology Specific Parameters For Non-Fossil Fuel Based Cogeneration Projects

46.

Technology Aspect

.-A project shall qualify as a non-fossil fuel based Co-generation project, if it is in accordance with the eligibility criteria as specified under regulation 4(4).

47.

Capital Cost

.-The normative capital cost for the non-fossil fuel based cogeneration projects shall be Rs. 445 Lakh/MW for the first year of Control Period (i.e. FY 2009-10), and shall be linked to indexation formula as outlined under regulation 48.

48.

#### Capital Cost Indexation Mechanism

.-The following indexation mechanism shall be applicable in case of non-fossil fuel based cogeneration projects for adjustments in capital cost with the changes in Wholesale Price Index for Steel and Electrical Machinery:

$$CC(n) = P\&M(n) * (1+F1+F2+F3)$$

$$P\&M(n) = P\&M(0) * (1+d(n))$$

$$D(n) = [a*\{(SI(n-1)/SI(0))-1\} + b*\{(EI(n-1)/EI(0))-1\}]/(a+b)$$

Where,-

CC(n) = Capital Cost for nth year

P&M(n) = Plant and Machinery Cost for nth year

Note. - P&M is to be computed by dividing the base capital cost (for the first year of the Control Period) by (1+F1+F2+F3), i.e., Rs. 445 Lakh per MW/1.33 = Rs. 335 Lakh per MW.

d(n) = Capital Cost escalation factor for year (n)

of Control Period

SI(n-1) = Average WPI Steel Index prevalent for fiscal year (n-1) of the Control Period

SI(0) = Average WPI Steel Index prevalent for calendar year (0) at the beginning of the Control Period, i.e., January 2008 to December 2008

EI(n-1) = Average WPI Electrical Machinery Index prevalent for calendar year (n-1) of the Control Period

EI(0) = Average WPI Electrical and Machinery Index prevalent for calendar year (0) at the beginning of the Control Period, i.e., January 2008 to December 2008

a = Constant to be determined by Commission from time to time,

(In default it is 0.7), for weightages to Steel Index

b = Constant to be determined by Commission from time to time,

(In default it is 0.3), for weightages to Electrical Machinery Index

F1 = Factor for Land and Civil Work (0.10)

F2 = Factor for Erection and Commissioning (0.09)

F3 = Factor for IDC and Financing Cost (0.14)

49.

#### Plant Load Factor

.- (1) For the purpose of determining fixed charge, the plant load factor for nonfossil fuel based cogeneration projects shall be computed on the basis of plant availability for number of operating days considering operations during crushing season and off-season as specified under clause (2) below and load factor of 92%.

(2)

The number of operating days for different States shall be as follows:

State

Operating Days

Plant Load Factor

(%)

Uttar Pradesh and Andhra Pradesh

120 days (crushing)+60 days (off-season)=180 days

Operating days

45%

Tamil Nadu and Maharashtra

180 days (crushing)+60 days (off-season)=240 days

Operating days

60%

Other States

150 days (crushing)+60 days(off-season)=210 days

Operating days

53%

50.

Auxiliary Consumption

.-The auxiliary power consumption factor shall be 8.5% for the computation of tariff.

51.

Station Heat Rate

.-The Station Heat Rate of 3600 kCal/kWh for power generation component alone shall be considered for computation of tariff for non-fossil fuel based Cogeneration projects.

52.

Calorific Value

.-The Gross Calorific Value for Bagasse shall be considered as 2250 kCal/kg. For the use of biomass fuels other than bagasse , calorific value as specified under regulation 43 shall be applicable.

53.

Fuel Cost

.- (1) The price of Bagasse shall be as specified in the table below and shall be linked to index formulae as outlined under regulation 54. Alternatively, for each subsequent year of the Control Period, the normative escalation factor of 5% per annum shall be applicable at the option of the project developer.

State

Bagasse Price

(Rs./MT)

Andhra Pradesh

899

Haryana

1411

Maharashtra

1123

Madhya Pradesh

809

Punjab

1398

Tamil Nadu

1243

Uttar Pradesh

1013

Other States

1163

(2)

For use of biomass other than bagasse in co-generation projects, the biomass prices as specified under regulation 44 shall be applicable.

54.

Fuel Price Indexation Mechanism

.- (1) In case of non-fossil fuel based cogeneration projects, the following indexing mechanism for adjustment of fuel prices for each year of operation will be applicable for determination of applicable variable charge component of tariff, in case developer wishes to opt for indexing mechanism:

$$P(n) = P(n-1) * \{a*(WPI(n)/WPI(n-1))+b*(1+IRC)(n-1)+c*(pd(n)/pd(n-1))\}$$

Where,-

$P(n)$  = Price per ton of Bagasse for the  $n$ th year to be considered for tariff determination

$P(n-1)$  = Price per ton of Bagasse for the  $(n-1)$ th year to be considered for tariff determination.  $P1$  shall be Biomass price for FY 2009-10 as specified under regulation 55.

$a$  = Factor representing fuel handling cost

$b$  = Factor representing fuel cost

$c$  = Factor representing transportation cost

$IRC(n-1)$  = Average Annual Inflation Rate for indexed energy charge component in case of captive coal mine source (in %) to be applicable for  $(n-1)$ th year, as may be specified by CERC for "Payment purpose" as per Compleitive Bidding Guidelines

$P_{dn}$  = Weighted average price of HSD for  $n$ th year.

$P_{dn-1}$  = Weighted average price of HSD for  $(n-1)$ th year.

$WPI_n$  = Whole sale price index for the month of April of  $n$ th year

$WPI_{n-1}$  = Wholesale price index for month of April of  $(n-1)$ th year.

Where  $a$ ,  $b$  &  $c$  will be specified by the Commission from time to time. I default, these factors shall be 0.2, 0.6 & 0.2 respectively.

(2)

Variable Charge for the  $n$ th year shall be determined as under :

i.e.  $VC_n = VC_1 \times (P_n/p_1)$

or  $VC_n = VC_1 \times (1.05)^{(n-1)}$  (optional)

$VC_1$  represents the Variable Charge based on bagasse Price  $P1$  for FY 2009-10 as specified under regulation 55 and shall be determined as under:

$VC_1 =$

Station Heat Rate

(SHR)

$\times$

1

$\times$

$P1$

Gross Calorific Value (GCV)

(1-Aux Consum.

Factor)

1000

55.

Operation and Maintenance Expenses

-(1) Normative O&M expenses during first year of the Control Period (i.e. FY 2009-10) shall be Rs. 13.35 Lakh per MW.

(2)

Normative O&M expenses allowed at the commencement of the Control Period (i.e. FY 2009-10) under these regulations shall be escalated at the rate of 5.72% per annum.

Chapter 7

Technology Specific Parameters For Solar Pv Power Project

56.

Technology Aspects

.-Norms for Solar Photovoltaic (PV) power under these regulations shall be applicable for grid connected PV systems

that directly convert solar energy into electricity and are based on the technologies such as crystalline silicon or thin film, etc., as may be approved by MNRE.

57.

#### Capital Cost

.-The normative capital cost for setting up Solar Photovoltaic Power Project shall be Rs. 1700 Lakh/MW for FY 2009-10: Provided that the Commission may deviate from above norm in case of project specific tariff determination in pursuance of regulation 7 and regulation 8.

58.

#### Capacity Utilisation Factor

.-The Capacity utilisation factor for Solar PV project shall be 19%:

Provided that the Commission may deviate from above norm in case of project specific tariff determination in pursuance of regulation 7 and regulation 8.

59.

#### Operation and Maintenance Expenses

.- (1) The O&M Expenses shall be Rs. 9 Lakhs/MW for the 1st year of operation.

(2)

Normative O&M expenses allowed at the commencement of the Control Period under these regulations shall be escalated at the rate of 5.72% per annum.

### Chapter 8

#### Technology Specific Parameters For Solar Thermal Power Project

60.

#### Technology Aspects

.-Norms for Solar thermal power under these regulations shall be applicable for Concentrated solar power (CSP) technologies, viz., line focusing or point focusing, as may be approved by MNRE, and uses direct sunlight, concentrating it several times to reach higher energy densities and thus higher temperatures whereby the heat generated is used to operate a conventional power cycle to generate electricity.

61.

#### Capital Cost

.-The normative capital cost for setting up Solar Thermal Power Project shall be Rs. 1300 Lakh/MW for FY 2009-10: Provided that the Commission may deviate from the above norm in case of project specific tariff determination in pursuance of regulation 7 and regulation 8.

62.

#### Capacity Utilisation Factor

.-The Capacity utilisation factor shall be 23%:

Provided that the Commission may deviate from the above norm in case of project specific tariff determination in pursuance of regulation 7 and regulation 8.

63.

#### Operation and Maintenance Expenses

.- (1) The O&M Expenses shall be Rs 13 Lakhs/MW for 1st year operation.

(2)

Normative O&M expenses allowed at the commencement of the Control Period under these regulations shall be escalated at the rate of 5.72% per annum.

64.

#### Auxiliary Consumption

.-The auxiliary consumption factor shall be 10%:

Provided that the Commission may deviate from the above norm in case of project specific tariff determination in pursuance of regulation 7 and regulation 8.

### Chapter 9

#### Miscellaneous

65.

#### Deviation from norms

.-Tariff for sale of electricity by the generating company may also be determined in deviation from the norms specified in these regulations subject to the conditions that the levellised tariff over the useful life of the project on the basis of the norms in deviation does not exceed the levellised tariff calculated on the basis of the norms specified in these regulations:

Provided that the reasons for deviation from the norms specified under these regulations shall be recorded in writing.

66.

Power to Relax

.-The Commission may by general or special order, for reasons to be recorded in writing, and after giving an opportunity of hearing to the parties likely to be affected may relax any of the provisions of these regulations on its own motion or on an application made before it by an interested person.

FORM-1.1

Form Template For (Wind Power Or Small Hydro Project Or Solar Pv/solar Thermal)

S. No.

AssumptionHead

Sub-Head

Sub-Head (2)

Unit

Parametervalues

1

Power Generation

Capacity

Installed Power Generation Capacity

MW

Capacity Utilization Factor

%

Commercial Operation Date

mm/yyyy

Useful Life

Years

2

Project Cost

Capital Cost/MW

Normative Capital Cost

Rs. Lakh/MW

Capital Cost

Rs. Lakh

Capital subsidy, if any

Rs. Lakh

Net Capital Cost

Rs. Lakh

3

Financial Assumptions

Tariff Period

Years

Debt: Equity

Debt

%

Equity

%

Total Debt Amount

Rs. Lacs

Total Equity Amount

Rs. Lacs

Debt component

Loan Amount

Rs. Lacs

Moratorium Period

Year Rs.

Repayment Period (includ Moratorium)

Year Rs.

Interest Rate

%

Equity component

Equity amount

Rs. Lacs

Return on Equity for first 10 yea Rs.

% p.a.

Return on Equity 11th year onwards

% p.a.

Discount Rate

%

Depreciation

Depreciation Rate for first 12 yea Rs.

%

Depreciation Rate 13th year onwards

%

Incentives

Generation Based Incentives, if any

Rs. L p.a.

Period for GBI

Year Rs.

4

Operation & Maintenance

Normative O&M expense

Rs. Lakh/MW

O&M expense per annum

Rs. Lakh

Escalation factor for O&M expense

%

5

Working Capital

O&M expense Months

Maintenance Spare

(% of O&M exepenses)

%

Receivables

Months

Interest on Working Capital

% p.a.

FORM-2.1

Form Template For (Biomass Power Or Non-Fossil Fuel Based Cogen) : Parameter Assumptions

S. No.

Assumption Head

Sub-Head

Sub-Head (2)

Unit

Parameter values

1

Power Generation

Capacity

Installed Power Generation Capacity

MW

Auxiliary Consumption factor

%

PLF (during stabilisation upto 6 months)

%

PLF (during 1st yr after stabilisation)

%

PLF (2nd yr onwards)

%

Commercial Operation Date

mm/yyyy

Useful Life

Years

2

Project Cost

Capital Cost/MW

Normative Capital Cost

Rs. Lakh/MW

Capital Cost

Rs. Lakh

Capital subsidy, if any

Rs. Lakh

Net Capital Cost

Rs. Lakh

3

Financial Assumptions

Tariff Period

Years

Debt: Equity

Debt

%

Equity

%

Total Debt Amount

Rs. Lacs

Total Equity Amount

Rs. Lacs

Debt component

Loan Amount

Rs. Lacs

Moratorium Period

years



Repayment Period (incl'd Moratorium)

years

Interest Rate

%

Equity component

Equity amount

Rs. Lacs

Return on Equity for first 10 years

% p.a

Return on Equity 11th year onwards

% p.a

Discount Rate

%

Depreciation

Depreciation Rate for first 12 years

%

Depreciation Rate 13th year onwards

%

Incentives

Generation Based Incentives, if any

Rs. L p.a.

Period for GBI

Years

4

Operation & Maintenance

Normative O&M expense

Rs. Lakh/MW

O&M expense per annum

Rs. Lakh

Escalation factor for O&M expense

%

5

Working Capital

O&M expense

Months

Maintenance Spare

(% of O&M expenses)

%

Receivables

Months

Biomass stock

Months

Interest on Working Capital

% p.a.

6

Fuel related assumptions

Station Heat Rate

during stabilisation

kcal/kWh

post stabilisation

kcal/kWh

Fuel types & mix

Biomass fuel type -1

%

Biomass fuel type -2

%

fossil Fuel (coal)

%

GCV of Biomass fuel type -1

kCal/kg

GCV of Biomass fuel type -2

kCal/kg

GCV of fossil Fuel (coal)

kCal/kg

Biomass Price (fuel type -1) : yr-1

Rs/MT

Biomass Price (fuel type -2) : yr-1

Rs/MT

Fossil fuel price (coal) : yr-1

Rs/MT

Fuel price escalation factor

% p.a.

FORM-1.2

Form Template For (Wind Power Or Small Hydro Project Or Solar Pv/solar Thermal) : Determination Of Tariff Components

Units Generation

Unit

Yr-1 Yr-2 Yr-3 Yr-4 Yr-5 Yr-6 Yr-7

Yr-8 Yr-9 Yr-10 Yr-11 Yr-12 Yr-13 Yr-14 Yr-15 Yr-16 Yr-17 Yr-18 Yr-19 Yr-20

Yr-21 Yr-22 Yr-23 Yr-24

Installed Capacity

MW

Net Generation

MU

Tariff Components (Fixed charge)

Unit

Yr-1 Yr-2 Yr-3 Yr-4 Yr-5 Yr-6 Yr-7 Yr-8 Yr-9

Yr-10 Yr-11 Yr-12 Yr-13 Yr-14 Yr-15 Yr-16 Yr-17 Yr-18 Yr-19 Yr-20 Yr-21 Yr-22

Yr-23 Yr-24

O&M Expenses

RsLakh

Depreciation

RsLakh

Interest on term loan

RsLakh

Interest on working Capital

RsLakh

Return on Equity

RsLakh

Total Fixed Cost

RsLakh

Per Unit Tariff components

|                                                                                                       |
|-------------------------------------------------------------------------------------------------------|
| Unit                                                                                                  |
| Yr-1 Yr-2 Yr-3 Yr-4 Yr-5 Yr-6 Yr-7 Yr-8 Yr-9                                                          |
| Yr-10 Yr-11 Yr-12 Yr-13 Yr-14 Yr-15 Yr-16 Yr-17 Yr-18 Yr-19 Yr-20 Yr-21 Yr-22                         |
| Yr-23 Yr-24                                                                                           |
| PU O&M Expenses                                                                                       |
| Rs/kWh                                                                                                |
| PU Depreciation                                                                                       |
| Rs/kWh                                                                                                |
| PU Interest on term loan                                                                              |
| Rs/kWh                                                                                                |
| PU Interest on working Capital                                                                        |
| Rs/kWh                                                                                                |
| PU Return on Equity                                                                                   |
| Rs/kWh                                                                                                |
| PU Tariff Components                                                                                  |
| Rs/kWh                                                                                                |
| LevellersTariff                                                                                       |
| Unit                                                                                                  |
| Yr-1 Yr-2 Yr-3 Yr-4 Yr-5 Yr-6 Yr-7 Yr-8 Yr-9                                                          |
| Yr-10 Yr-11 Yr-12 Yr-13 Yr-14 Yr-15 Yr-16 Yr-17 Yr-18 Yr-19 Yr-20 Yr-21 Yr-22                         |
| Yr-23 Yr-24                                                                                           |
| Discount Factors                                                                                      |
| Discounted Tariff components                                                                          |
| Rs/kWh                                                                                                |
| LevellersTariff                                                                                       |
| Rs/kWh                                                                                                |
| FORM-2.2                                                                                              |
| Form Template For (Biomass Power Or Non-Fossil Fuel Based Cogen) : Determination Of Tariff Components |
| Units Generation                                                                                      |
| Unit                                                                                                  |
| Yr-1 Yr-2 Yr-3 Yr-4 Yr-5 Yr-6 Yr-7                                                                    |
| Yr-8 Yr-9 Yr-10 Yr-11 Yr-12 Yr-13 Yr-14 Yr-15 Yr-16 Yr-17 Yr-18 Yr-19 Yr-20                           |
| Yr-21 Yr-22 Yr-23 Yr-24 Yr-25                                                                         |
| Installed Capacity                                                                                    |
| MW                                                                                                    |
| Net Generation                                                                                        |
| MU                                                                                                    |
| Tariff Components (Fixed charge)                                                                      |
| Unit                                                                                                  |
| Yr-1 Yr-2 Yr-3 Yr-4 Yr-5 Yr-6 Yr-7                                                                    |
| Yr-8 Yr-9 Yr-10 Yr-11 Yr-12 Yr-13 Yr-14 Yr-15 Yr-16 Yr-17 Yr-18 Yr-19 Yr-20                           |
| Yr-21 Yr-22 Yr-23 Yr-24 Yr-25                                                                         |
| O&M Expenses                                                                                          |
| RsLakh                                                                                                |
| Depreciation                                                                                          |
| RsLakh                                                                                                |
| Interest on term loan                                                                                 |
| RsLakh                                                                                                |
| Interest on working Capital                                                                           |
| RsLakh                                                                                                |
| Return on Equity                                                                                      |

|                                                                             |
|-----------------------------------------------------------------------------|
| RsLakh                                                                      |
| Total Fixed Cost                                                            |
| RsLakh                                                                      |
| Tariff Components (Variable charge)                                         |
| Unit                                                                        |
| Yr-1 Yr-2 Yr-3 Yr-4 Yr-5 Yr-6 Yr-7                                          |
| Yr-8 Yr-9 Yr-10 Yr-11 Yr-12 Yr-13 Yr-14 Yr-15 Yr-16 Yr-17 Yr-18 Yr-19 Yr-20 |
| Yr-21 Yr-22 Yr-23 Yr-24 Yr-25                                               |
| Biomass fuel type-1                                                         |
| RsLakh                                                                      |
| Biomass fuel type-2                                                         |
| RsLakh                                                                      |
| Fossil fuel (coal)                                                          |
| RsLakh                                                                      |
| Sub-total (Fuel Costs)                                                      |
| RsLakh                                                                      |
| Fuel cost allocable to power                                                |
| %                                                                           |
| Total Fuel Costs                                                            |
| RsLakh                                                                      |
| Per Unit Tariff components (fixed)                                          |
| Unit                                                                        |
| Yr-1 Yr-2 Yr-3 Yr-4 Yr-5 Yr-6 Yr-7                                          |
| Yr-8 Yr-9 Yr-10 Yr-11 Yr-12 Yr-13 Yr-14 Yr-15 Yr-16 Yr-17 Yr-18 Yr-19 Yr-20 |
| Yr-21 Yr-22 Yr-23 Yr-24 Yr-25                                               |
| PU O&M Expenses                                                             |
| Rs/kWh                                                                      |
| PU Depreciation                                                             |
| Rs/kWh                                                                      |
| PU Interest on term loan                                                    |
| Rs/kWh                                                                      |
| PU Interest on working Capital                                              |
| Rs/kWh                                                                      |
| PU Return on Equity                                                         |
| Rs/kWh                                                                      |
| PU Tariff Components (Fixed)                                                |
| Rs/kWh                                                                      |
| PU Tariff Components (Variable)                                             |
| Rs/kWh                                                                      |
| PU Tariff Components (Total)                                                |
| Rs/kWh                                                                      |
| LevellingTariff                                                             |
| Unit                                                                        |
| Yr-1 Yr-2 Yr-3 Yr-4 Yr-5 Yr-6 Yr-7                                          |
| Yr-8 Yr-9 Yr-10 Yr-11 Yr-12 Yr-13 Yr-14 Yr-15 Yr-16 Yr-17 Yr-18 Yr-19 Yr-20 |
| Yr-21 Yr-22 Yr-23 Yr-24 Yr-25                                               |
| Discount Factors                                                            |
| Discounted Tariff components (fixed)                                        |
| Rs/kWh                                                                      |
| Discounted Tariff components                                                |
| (variable)                                                                  |

Rs/kWh

Discounted Tariff components (total)

Rs/kWh

LevellingTariff (fixed)

Rs/kWh

LevellingTariff (variable)

Rs/kWh

LevellingTariff (total)

Rs/kWh

State-wise Wind Power Density Map

Related AI tags, queries and research notes

Translation